



Project Progress Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	25
Problem understanding & Analysis	7					
Implementation	8					
Report writing	10					
Total obtained mark						
Comments						

Semester: Spring 2026

Student Name: Sudipto Biswas, Shah Shazid Askary, Masud Hassan Sakib

Student ID: 0242310005341163, 0242310005341201, 0242310005341046

Batch: 40

Section: E1

Course Code: SE331

Course Name: Software Engineering

Design Capstone Project

Course Teacher Name: Rahat Uddin Azad

Designation: Lecturer

Submission Date: 30.03.2026

Project Progress Report

1. Implemented Features

Authentication & User Management

Phone number-based OTP login & registration: **Completed**

OTP verification screen with 5-minute expiry & 5 attempt limit: **Completed**

JWT token issuance (access + refresh): **Completed**

Business profile setup (name, type, address): **Completed**

Forgot password / Reset password flow: **Completed**

Language switch (Bengali ↔ English) without restart: **Completed**

Staff member management (merchant → staff linking): **Partially Completed**

Dashboard & Summary

Real-time today's summary: Sales, Purchases, Expenses, Receivable, Payable, Gross Profit: **Completed**

5 most recent transactions on dashboard: **Completed**

Total receivable/payable with contact counts: **Completed**

Online/offline sync status indicator & pending sync count: **Completed**

Quick action buttons (বিক্রয়, ক্রয়, খরচ, etc.): **Completed**

Dashboard real-time update on new transaction: **Completed**

Transaction Management

7 transaction types: Sale, Purchase, Expense, Income, Payment In, Payment Out, Return: **Completed**

Full transaction form: amount, paid amount, due, contact, date, payment method, notes: **Completed**

Payment methods: Cash, bKash, Nagad, Rocket, Bank, Credit: **Completed**

Auto-generated unique reference numbers: **Completed**

Filter transactions by type, date range, contact: **Completed**

Search transactions by contact name / reference number: **Completed**

Soft deletion of transactions for sync integrity: **Completed**

Transaction line items (itemized sales/purchases): **Partially Completed**

Image attachment (receipt/invoice) per transaction: **Not Completed**

Contact Management

3 contact types: Customer (গ্রাহক), Supplier (সরবরাহকারী), Both: **Completed**

Contact fields: name, phone, email, address, type, balance: **Completed**

Auto-calculated & updated contact balance from transactions: **Completed**

Debtor / Creditor flagging with color indicators: **Completed**

Transaction history per contact (ContactDetailScreen): **Completed**

Optional credit limit per contact: **Partially Completed**

Inventory Management

Product catalog: name, SKU, barcode, description, category: **Completed**

Purchase price, selling price, wholesale price tracking: **Completed**

Stock quantity tracking with configurable unit (pcs/kg/litre): **Completed**

Stock auto-update on sale (decrease) / purchase (increase): **Completed**

Low-stock alerts when stock falls below threshold: **Completed**

Stock adjustment log (addition, subtraction, damage, correction):
Completed

Profit margin calculation per product: **Completed**

Payment Integration (MFS)

Payment logging for bKash, Nagad, Rocket, Bank Transfer: **Completed**

Payment fields: gateway, amount, gateway TX ID, sender/receiver numbers, status: **Completed**

Payment status tracking: Pending, Success, Failed, Cancelled, Refunded: **Completed**

Merchant wallet: total received / sent tracking: **Partially Completed**

QR code-based payment identification: **Not Completed**

Offline-First Sync

Full offline operation using Zustand + AsyncStorage: **Completed**

Offline change queue — auto-syncs when connectivity restored: **Completed**

Sync queue tracks action type (create/update/delete) per entity: **Completed**

Pending sync count and last sync timestamp on dashboard: **Completed Reports &**

Analytics

Profit/loss overview with date range filtering: **Completed**

Category-wise expense breakdown: **Completed**

Sales & purchase trend charts: **Completed**

Contact-level financial summaries: **Completed**

Localization (i18n)

Full Bengali (বাংলা) + English UI using i18next: **Completed**

Language switch persisted in user profile without app restart: **Completed**

All UI strings (buttons, labels, error messages) fully translated: **Completed**

Currency displayed as BDT (৳) throughout: **Completed**

2. Mapping to SRS Requirements

Map each implemented feature to the corresponding SRS requirement Ensure clear alignment between your **SRS** and implementation

SRS ID	Requirement	Implementation Status
FR-AUTH-01	Registration via phone number with OTP	Completed
FR-AUTH-02	6-digit OTP to phone for login	Completed
FR-AUTH-03	OTP expiry (5 min) & max 5 attempts	Completed
FR-AUTH-04	JWT access (1-day) + refresh (30-day) tokens	Completed
FR-AUTH-05	Business profile setup	Completed
FR-AUTH-06	Staff member management	Partially Completed
FR-AUTH-07	Language switch (Bengali ↔ English)	Completed
FR-TXN-01	7 transaction types	Completed
FR-TXN-02	Full transaction fields	Completed
FR-TXN-03	Auto-generated reference numbers	Completed
FR-TXN-04	Payment methods (Cash, bKash, Nagad...)	Completed
FR-TXN-05	Auto-update today's dashboard summary	Completed
FR-TXN-06/07	Filter & search transactions	Completed
FR-TXN-08	Transaction line items	Partially Completed
FR-TXN-09	Receipt image attachment	Not Completed
FR-TXN-10	Soft deletion for sync integrity	Completed
FR-DASH-01 05	Full dashboard summary & sync status	Completed

FR-CON-01 ₀₆	Contact management & balance tracking	Completed
FR-INV-01 ₀₇	Full inventory feature set	Completed
FR-PAY-01 ₀₃	MFS payment logging & status	Completed
FR-PAY-03	QR code payment identification	Not Completed
FR-PAY-04	Merchant wallet	Partially Completed

SRS ID	Requirement	Implementation Status
FR-SYNC-01 ₀ 4	Offline-first & auto sync	Completed

3. Testing Plan

Test Cases (Table Format)

Test ID	Test steps	Test Data	Expected Output	Test Status
TC AUTH-01	1. Open app → Enter phone 01700000001 → Tap “ওটপ পাঠান” → Enter 6-digit OTP → Submit	Phone: 01700000001, OTP: 123456	JWT tokens issued; user navigated to Dashboard	Pass
TC AUTH-02	1. Enter wrong OTP 5 times consecutively	OTP: 000000 (wrong) × 5	Account locked; “বেশি চেষ্টা করা হয়েছে” error shown	Pass
TC AUTH-03	1. Login → Go to MoreScreen → Tap language toggle (EN ↔ BN) → Verify all labels change	Language: Bengali → English	All UI text switches to English without app restart	Pass
TC TXN-01	1. Tap “টি বক্রয়” quick button → Enter amount ৳500 → Select contact → Tap Save	Amount: 500, Contact: “Rahim Traders”	Transaction saved; Dashboard total sales += ৳500; Contact balance updated	Pass
TC TXN-02	1. Try to add transaction with empty amount field → Tap Save	Amount: (empty)	Validation error “সঠিক পরিমাণ নির্দেয়” shown; transaction not saved	Pass
TC TXN-03	1. Add transaction → Open TransactionsScreen → Search by contact name	Search: “Rahim”	Only transactions linked to “Rahim Traders” shown	Pass

TC TXN-04	1. Filter transactions by type “Expense” and date range “This Month”	Type: Expense, Date: March 2026	Only expense transactions for March 2026 listed	Pass
--------------	--	------------------------------------	---	------

Test ID	Test steps	Test Data	Expected Output	Test Status
TC CON-01	1. Add contact (Customer) → Record ₺1000 sale to them → View receivable (they owe us)	Contact: “Karim Store”, Sale: ₺1000	Contact balance shows ₺1000	Pass
TC CON-02	1. Record “Payment In” ₺500 from same contact	Payment In: ₺500 from “Karim Store”	Contact balance reduces to ₺500	Pass
TC INV-01	1. Add product (stock: 10) → Record sale of 3 units → Check inventory	Product: “Rice 1kg”, Initial stock: 10, Sold: 3	Stock decreases to 7	Pass
TC INV-02	1. Set low-stock threshold to 5 → Reduce stock to 4 units	Threshold: 5, Current stock: 4	Low-stock alert notification triggered	Pass
TC SYNC-01	1. Turn off WiFi → Add a sale transaction → Turn on WiFi → Wait 5 seconds	Offline sale: ₺300	Transaction queued offline; auto-synced when reconnected; pending count → 0	Pass
TC DASH-01	1. Open Dashboard → Verify all 6 summary values are correct	Multiple prior transactions	Correct totals for Sales, Purchases, Expenses, Receivable, Payable, Profit	Pass

Include

Functional test cases

Basic validation cases

Test ID	Test steps	Test Data	Expected Output	Test Status
TC VAL - 01	Enter negative amount in transaction form	Amount: -500	Validation error; form not submitted	Pass
TC VAL - 02	Enter invalid phone number format during registration	Phone: “abc123”	“সঠিক ফো ন নম্বর দি ন” error shown	Pass
TC VAL - 03	Submit add contact with empty name field	Name: (empty)	Required field error shown; contact not saved	Pass
TC VAL - 04	Access API endpoints without JWT token	Request without Authorization header	401 Unauthorized response from backend	Pass
TC VAL - 05	Add product with stock = 0 and attempt sale	Stock: 0, Sale quantity: 1	Warning “স্টক নে ই” shown to user	Pass

4. Individual Contribution (Mandatory)

Each team member must clearly mention:

Modules / features worked on

Files / components contributed

Key challenges faced

What you learned during development

This section is **very important** and will be used for **individual evaluation**

Sudipto (Lead Developer · Backend & Architecture)

Modules / Features Worked On:

Overall system architecture & SRS document

Django backend setup (apps, models, serializers, views)

Authentication: OTP system, JWT token flow

Transaction Management API (CRUD + filtering)

Offline sync engine (syncStore.ts + /api/sync/)

Database design (PostgreSQL)

Railway deployment & Procfile, docker-compose

Files / Components Contributed:

backend/apps/users/ (models, serializers, views)

backend/apps/transactions/ (full CRUD)

mobile/src/store/syncStore.ts

mobile/src/store/transactionStore.ts

SRS_TakaTracker.md , railway.toml , docker-compose.yml

Key Challenges Faced:

Implementing reliable offline-first sync without data conflicts when the same record was modified both locally and server-side simultaneously. Managing the team members, integrating code and ideas while maintaining the flow of work.

Learned during development:

How to lead and contribute in the team. Designing conflict-resolution strategies using local UUIDs, tombstoning (soft delete), and queue-based sync. Deepened understanding of Django ORM signal handling and Celery task queues for async operations.

Shazid (Developer · Contacts, Payments & i18n)**Modules / Features Worked On:**

Contact management: Add, List, Detail screens

Contact balance auto-calculation from transactions

Payment logging screen (MFS: bKash, Nagad, Rocket)

Backend: contacts app, payments app models & APIs

Full i18n setup: Bengali + English (en.json, bn.json)

MoreScreen (settings, language toggle, logout)

Database design ERD

Notification store & alerts system

Files / Components Contributed:

mobile/src/screens/contacts/ (all 3 screens)

mobile/src/screens/payments/ & notifications/

mobile/src/store/contactStore.ts

mobile/src/i18n/locales/en.json & bn.json

mobile/src/screens/MoreScreen.tsx

backend/apps/contacts/ & backend/apps/payments/

Key Challenges Faced:

Ensuring the contact balance was always consistent — balances had to reflect every transaction type (sale, payment_in, purchase, payment_out) correctly, including edge cases like returns and partial payments.

Learned during development:

How to implement reactive i18n with i18next in React Native so language changes

propagate instantly. Importance of atomic balance update operations to prevent race conditions in the store.

Masud (Frontend Developer · Mobile UI & UX)

Modules / Features Worked On:

- Dashboard Screen design & summary cards
- Add/Edit Transaction Screen (full form)
- Transactions List Screen with filters & search
- Inventory Screen (product catalog, stock management)
- Reports Screen (charts, profit/loss analysis)
- Navigation structure (React Navigation stack & tabs)
- Custom components (badges, summary cards, buttons)

Files / Components Contributed:

- mobile/src/screens/dashboard/DashboardScreen.tsx
- mobile/src/screens/transactions/ (both screens)
- mobile/src/screens/inventory/InventoryScreen.tsx
- mobile/src/screens/reports/ReportsScreen.tsx
- mobile/src/navigation/ (stack & tab navigators)
- mobile/src/components/ (reusable UI components)

Key Challenges Faced:

Managing complex form state in AddTransactionScreen with dynamic field visibility depending on transaction type, while keeping smooth 60fps animations.

Learned during development:

Advanced React Native performance optimization using useMemo, useCallback, and FlatList virtualization. Gained strong understanding of Zustand for global state and how to keep UI reactive without unnecessary re-renders.